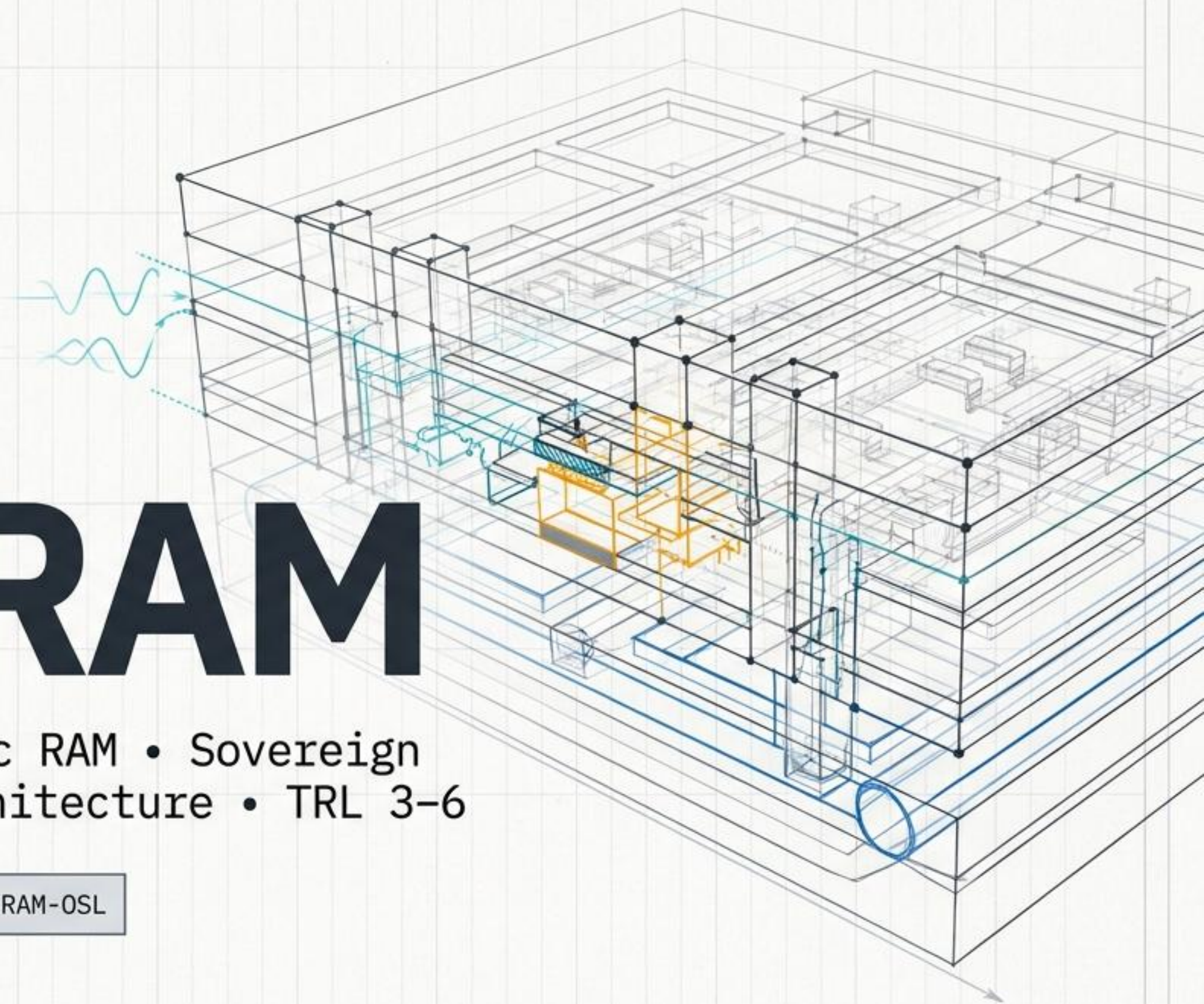


PCME-RAM

Phonon-Coupled Magnetolectric RAM • Sovereign
AI Research • Open Origin Architecture • TRL 3-6

Authored by Emanuel Schaaf • Released under PCME-RAM-OSL



4. Elastic Packaging

Significant increase in alignment tolerances achieved through elastic coupling layers (h-BN epoxy).

Legacy TSV

PCME-RAM
Architecture

1. Vertical Acoustic Data Transfer

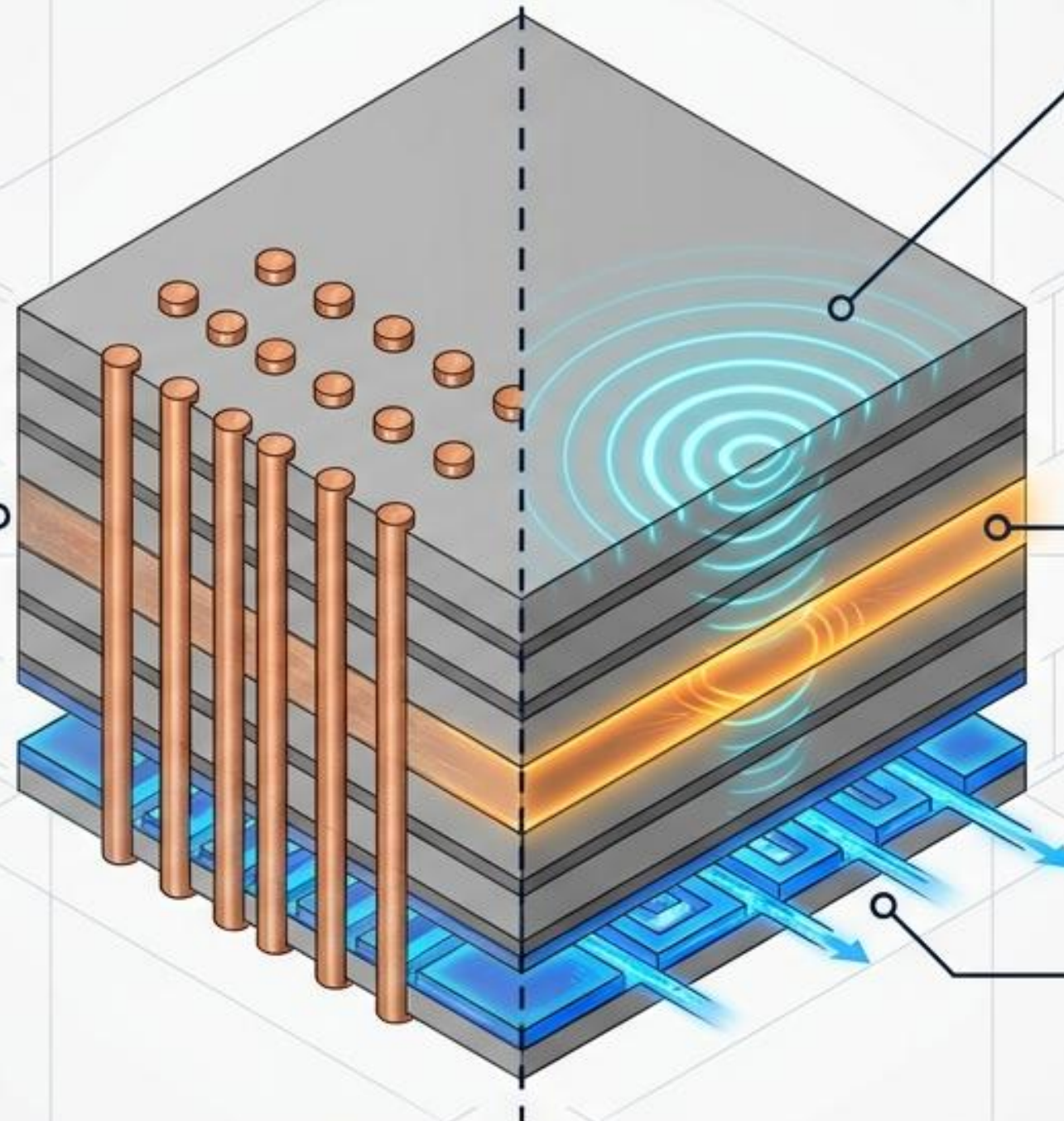
Partial substitution of signal-carrying copper TSVs with highly directional acoustic wave fields.

2. Strain-Switched Memory

Magnetoelectric heterostructures switching purely via strain control, eliminating Joule heating.

3. Integrated Microfluidics

Direct heat dissipation utilizing microscopic capillary effects etched into the wafer stack.



WARNING: Performance metrics reflect theoretical FDM capacity models and research targets defining the TRL 3-6 pathway, pending industrial validation.

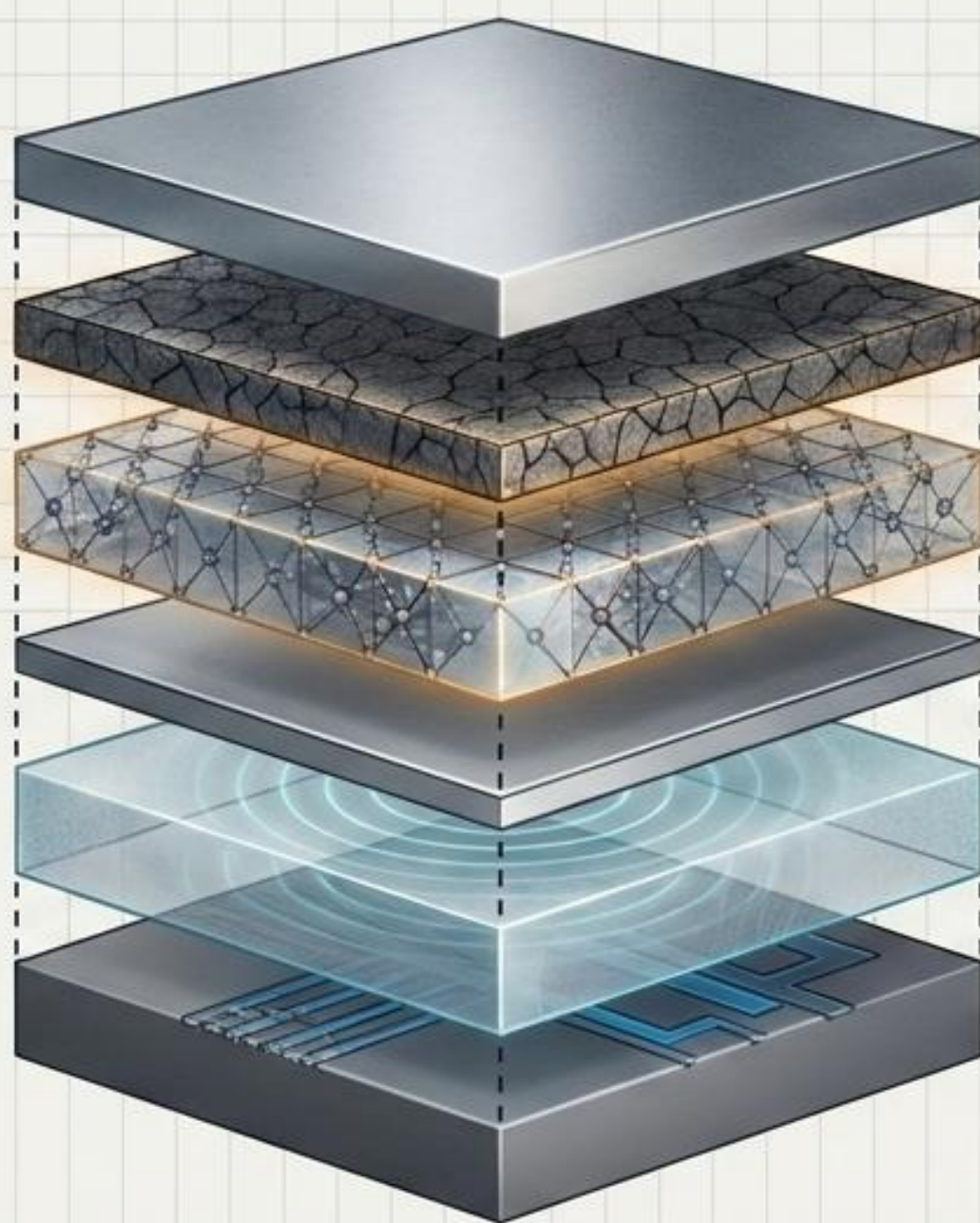
The Paradigm Shift Matrix

Architectural evaluation against industry standard capabilities.

	HBM4E (Industry Standard)	PCME-RAM (Research Goal)
Memory Type	DRAM	MeRAM
Volatility / Refresh	Requires Refresh	Non-Volatile
Cooling Paradigm	Conventional	Integrated Microfluidics
Bandwidth Target	3.6–4.0 TB/s	4.8 TB/s (FDM Capacity Model)
Latency	~12 ns	<2.5 ns (Modeling Assumption)

The Master Blueprint

Chemical material system and functional layer definition.



Ru (20 nm)

Top-Electrode: Adhesion promoter and electrical contact.

CoFe₂O₄ (15 nm)

Magnetostrictive layer: Free of rare-earth elements.

BaTiO₃ (50 nm)

Piezoelectric actuator: Lead-free.

SrRuO₃ (30 nm)

Bottom-Electrode / Buffer layer.

AlN (80 nm)

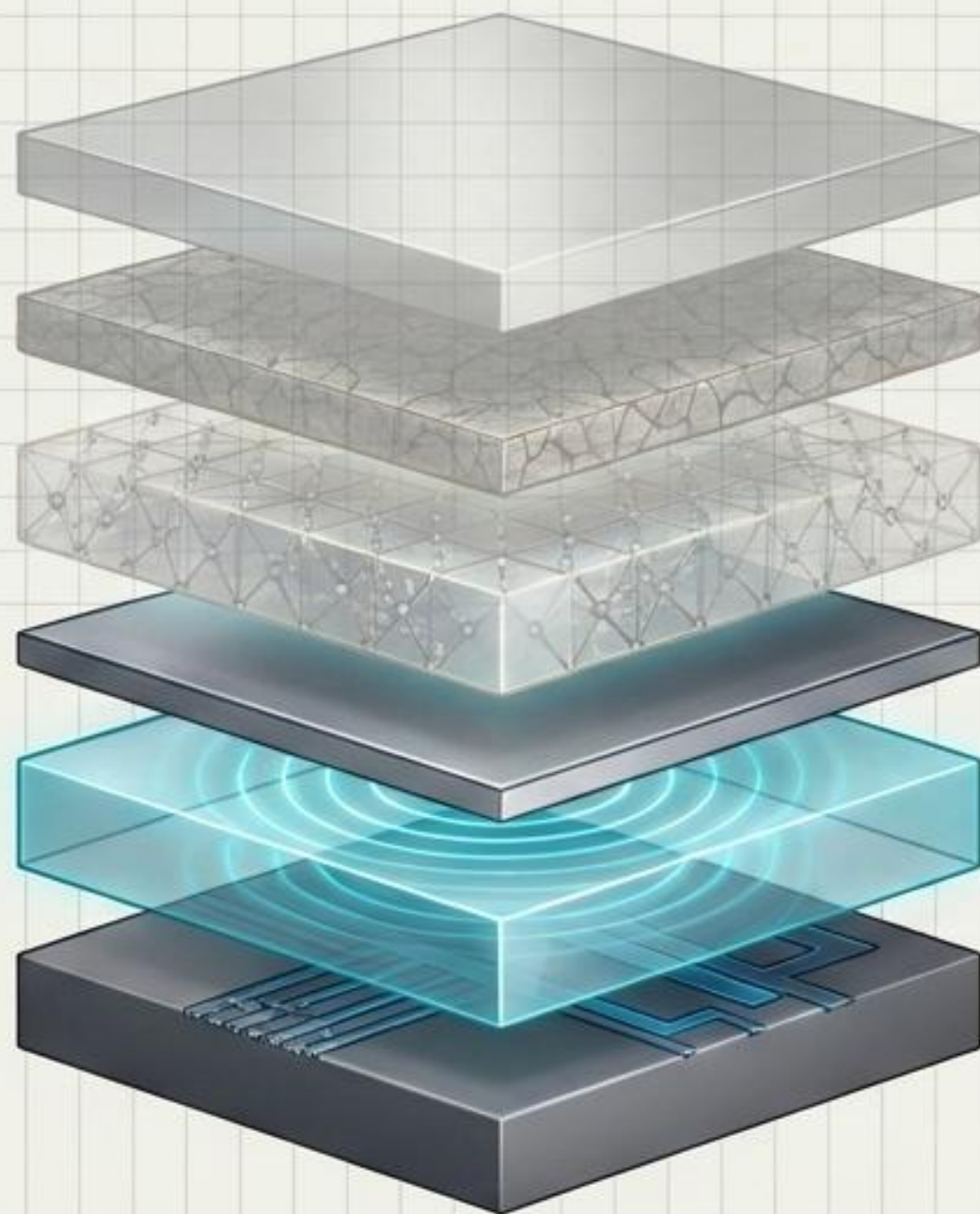
Acoustic transducer: MEMS-compatible.

Si(100)

Highly doped silicon substrate.

Physics Pillar I: Acoustic Wave Data Transfer

Elastodynamics and acoustic phonon propagation models.



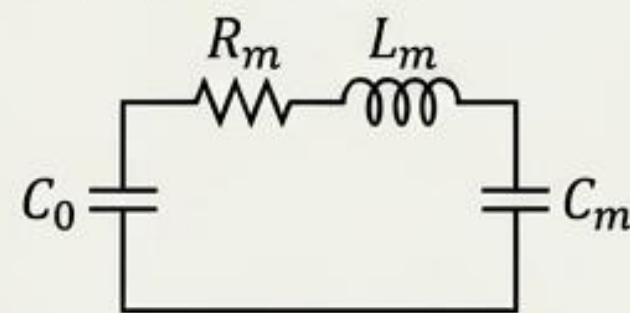
Elastodynamics

Elastic Wave Equation: $\rho_{Si} \frac{\partial^2 u_i}{\partial t^2} = c_{ijkl} \frac{\partial^2 u_k}{\partial x_j \partial x_l}$

Phase velocity (v_{Si}) ≈ 8433 m/s

Transit time (t_{prop}) ≈ 11.86 ps

Transducer Model



Butterworth-Van-Dyke (BVD)
Equivalent:

$$Z(\omega) = \frac{1}{j\omega C_0 + \frac{1}{R_m + j\omega L_m} + \frac{1}{j\omega C_m}}$$

CRITICAL ARCHITECTURAL BOTTLENECKS:

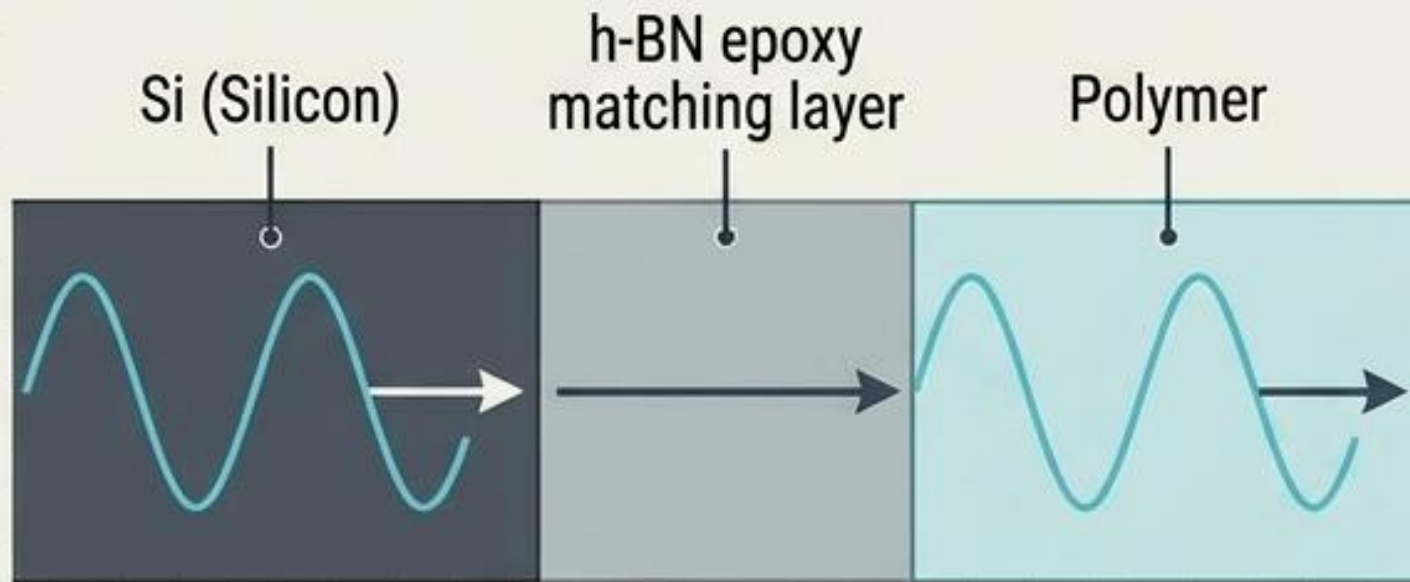
1. Acoustic Attenuation: $A(x) = A_0 e^{-\alpha x}$
2. Signal Crosstalk: $X_T = 20 \log_{10}(V_{\text{victim}}/V_{\text{aggressor}})$

Engineering the Waves: Metamaterials

Solving attenuation and crosstalk via optical physics equivalents.

Quarter-Wave Transformer

Solving Exponential Attenuation



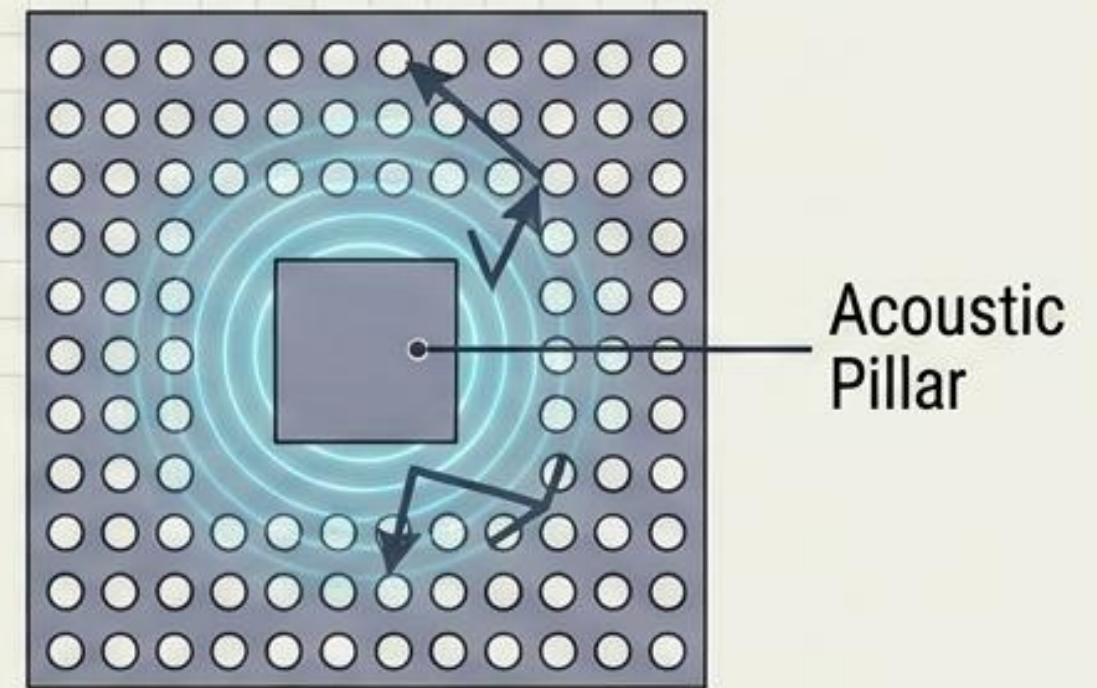
Impedance Matching: $Z_{\text{match}} = \sqrt{Z_{\text{Si}} \cdot Z_{\text{Polymer}}}$

Thickness Tuning (Destructive Interference):

$$d_{\text{match}} = \frac{\lambda}{4} = \frac{v_{\text{match}}}{4f_c}$$

Phononic Bandgap Engineering

Solving Signal Crosstalk (X_T)

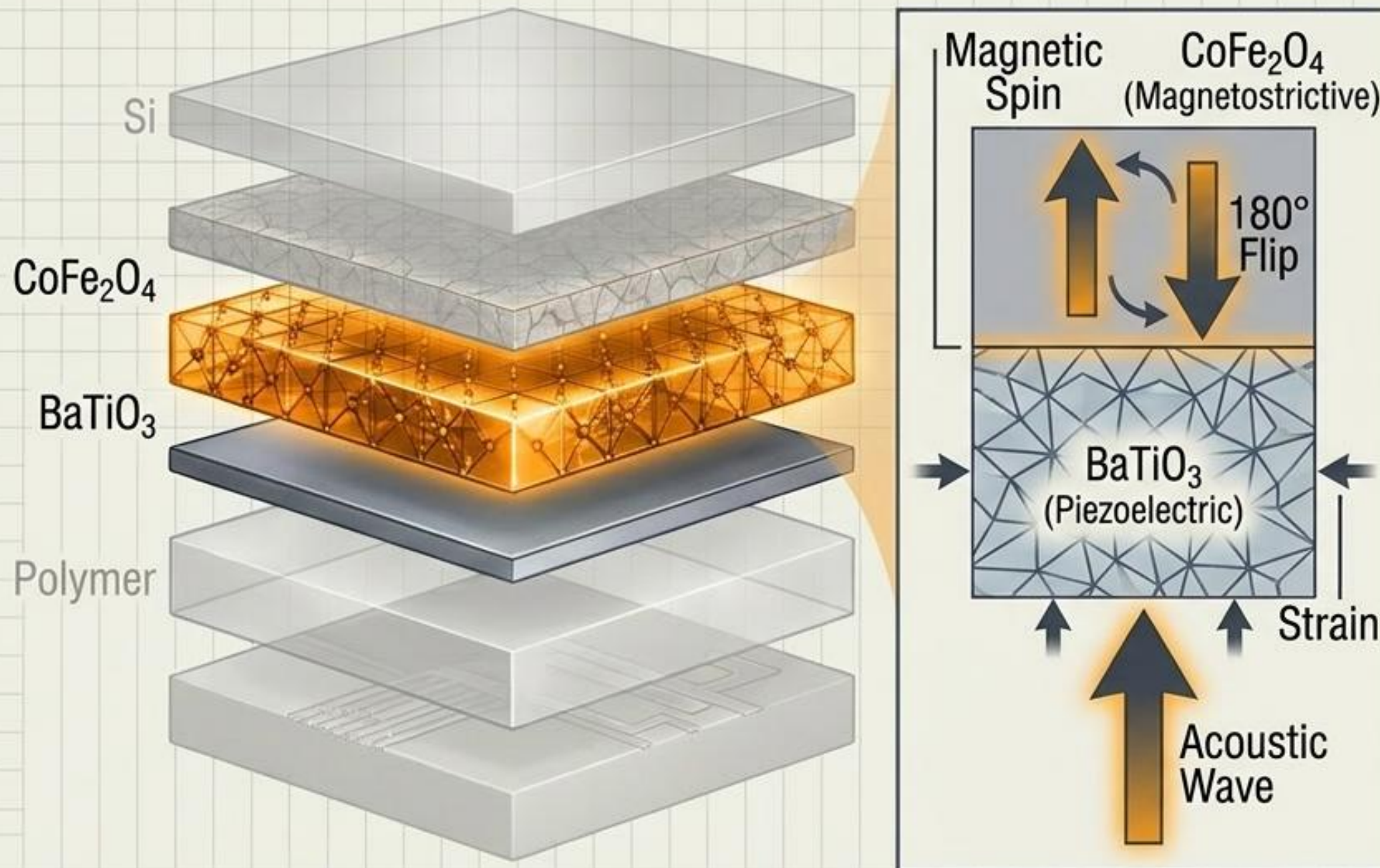


Bragg Scattering Condition: $a = \frac{v_{\text{transverse}}}{2f}$

Transverse leakage waves are exponentially dampened, dropping X_T below the thermal noise floor.

Physics Pillar II: Magnetoelectric Switching

Eliminating Joule heating via strain-mediated control



Magnetization Dynamics:

Landau-Lifshitz-Gilbert (LLG) Equation:

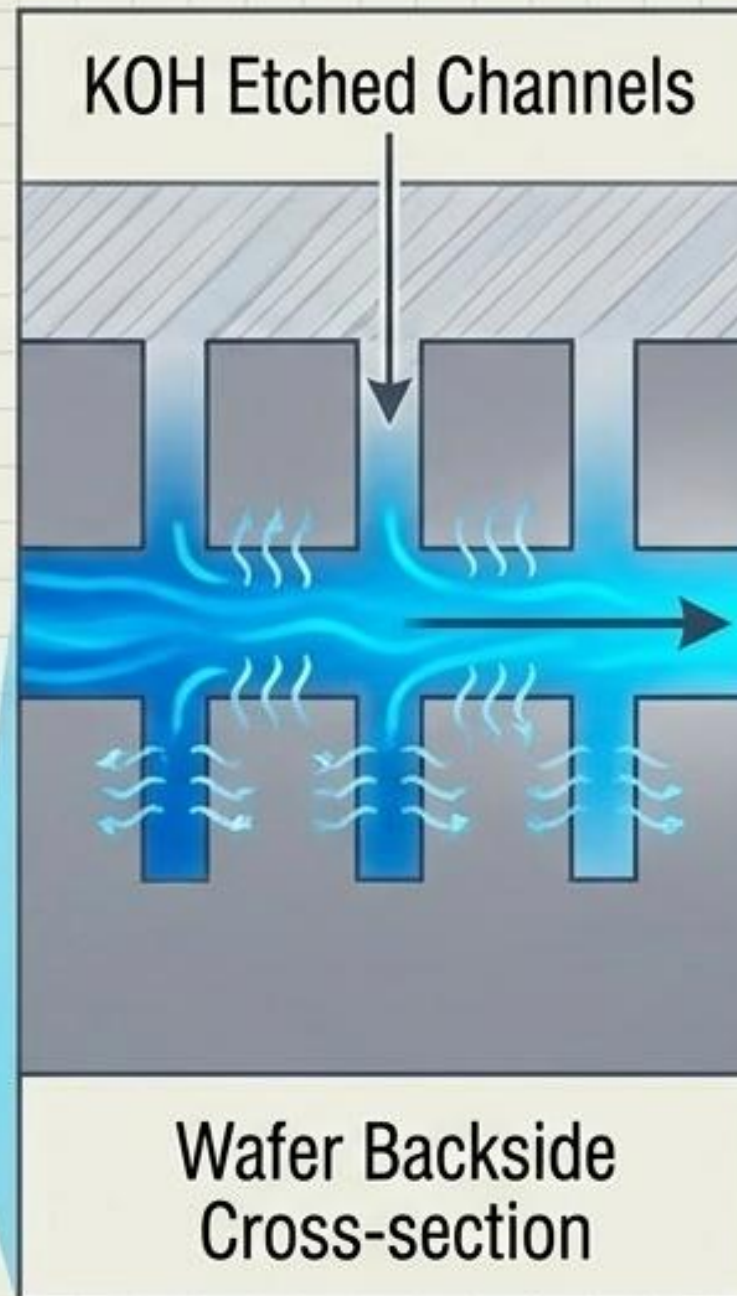
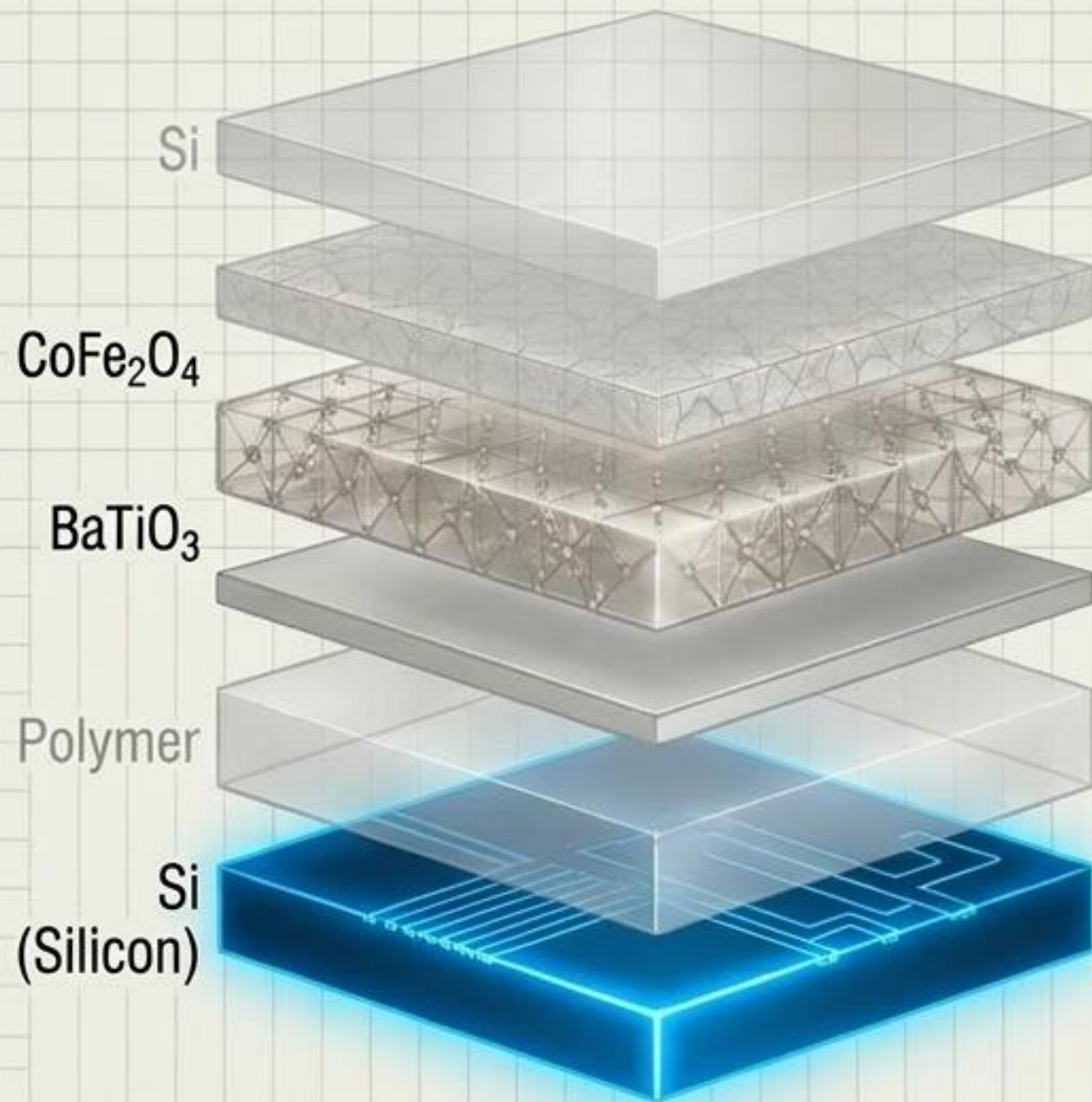
$$\frac{d\mathbf{M}}{dt} = -\gamma(\mathbf{M} \times \mathbf{H}_{\text{eff}}) + \frac{\alpha}{M_s} \left(\mathbf{M} \times \frac{d\mathbf{M}}{dt} \right)$$

FUNCTIONAL DECOUPLING

By utilizing strain-mediated magnetization control, state changes occur **purely mechanically**. This structurally **eliminates Joule heating** during the memory write-cycle.

Physics Pillar III: Thermodynamics

Integrated microfluidic capillary engine.



Capillary Engine:

Young-Laplace Equation:

$$\Delta P_{\text{cap}} = \frac{2\gamma \cos(\theta_c)}{r_{\text{eff}}} \approx 10.71 \text{ kPa}$$

Transient Thermal FEM:

$$\rho c_p \frac{\partial T}{\partial t} = \nabla(k \nabla T) + Q$$

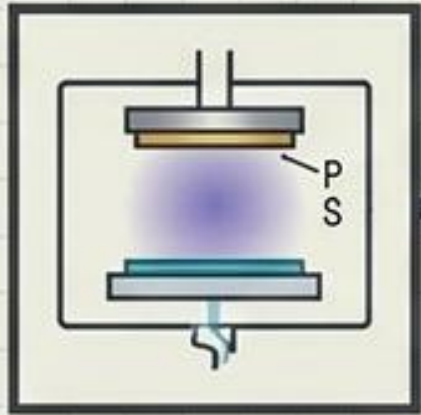
SIMULATION TARGET

Theoretical cooling capacity:
>520 W/cm²

Note: Validation of dry-out parameters pending.

Fabrication Protocol

TRL 3 / 4 industrial manufacturing flow.



1. Transducer

Reactive magnetron sputtering of AlN at 350°C.



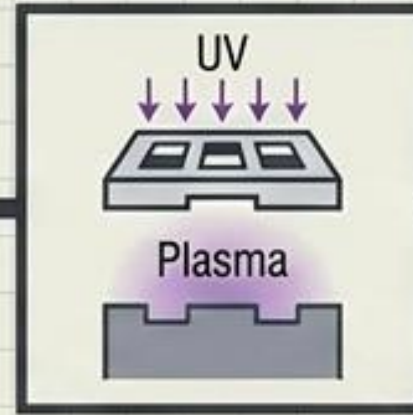
2. Bottom-Electrode

Deposition of SrRuO_3 via Pulsed Laser Deposition (PLD).



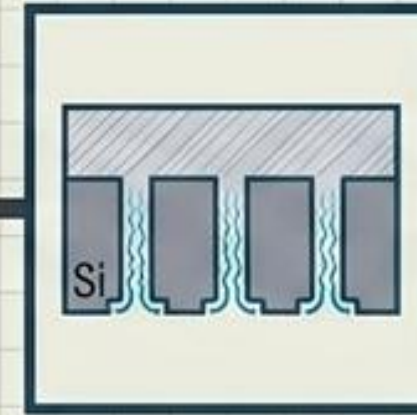
3. Heterostructure

BaTiO_3 / CoFe_2O_4 combined deposition under strict oxygen control.



4. Patterning

DUV lithography and ion beam etching (IBE).



5. Microchannels

ALD passivation and wet-chemical KOH etching (wafer backside).



6. Coupling

Spin-coating of h-BN epoxy matrix followed by UV curing.

Economics & Yield Models

Cost-reduction via elimination of extreme Cu-Cu hybrid bonding.

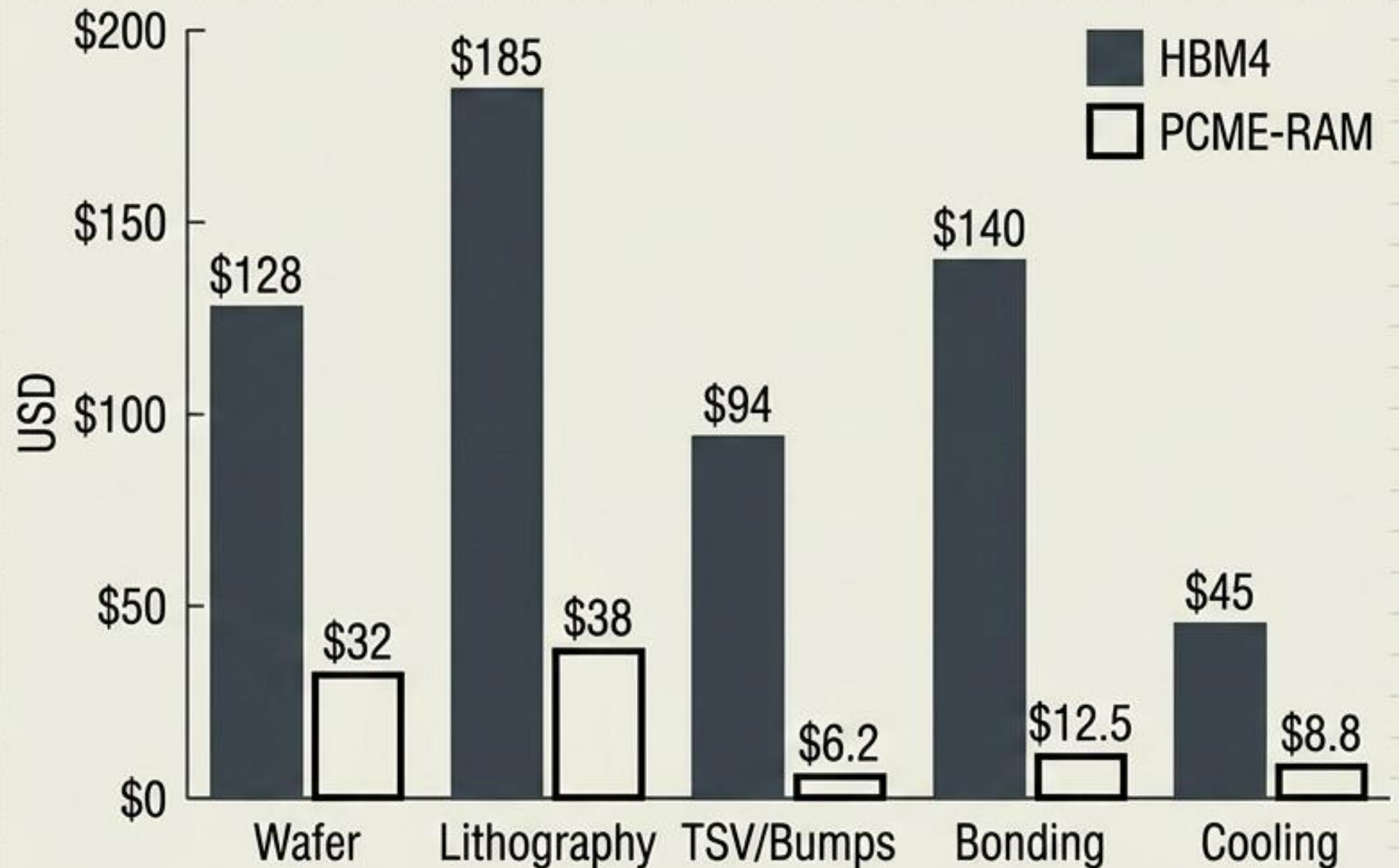
Yield Models

Theoretical Yield Mathematics:

Poisson Model: $Y = e^{-DA}$

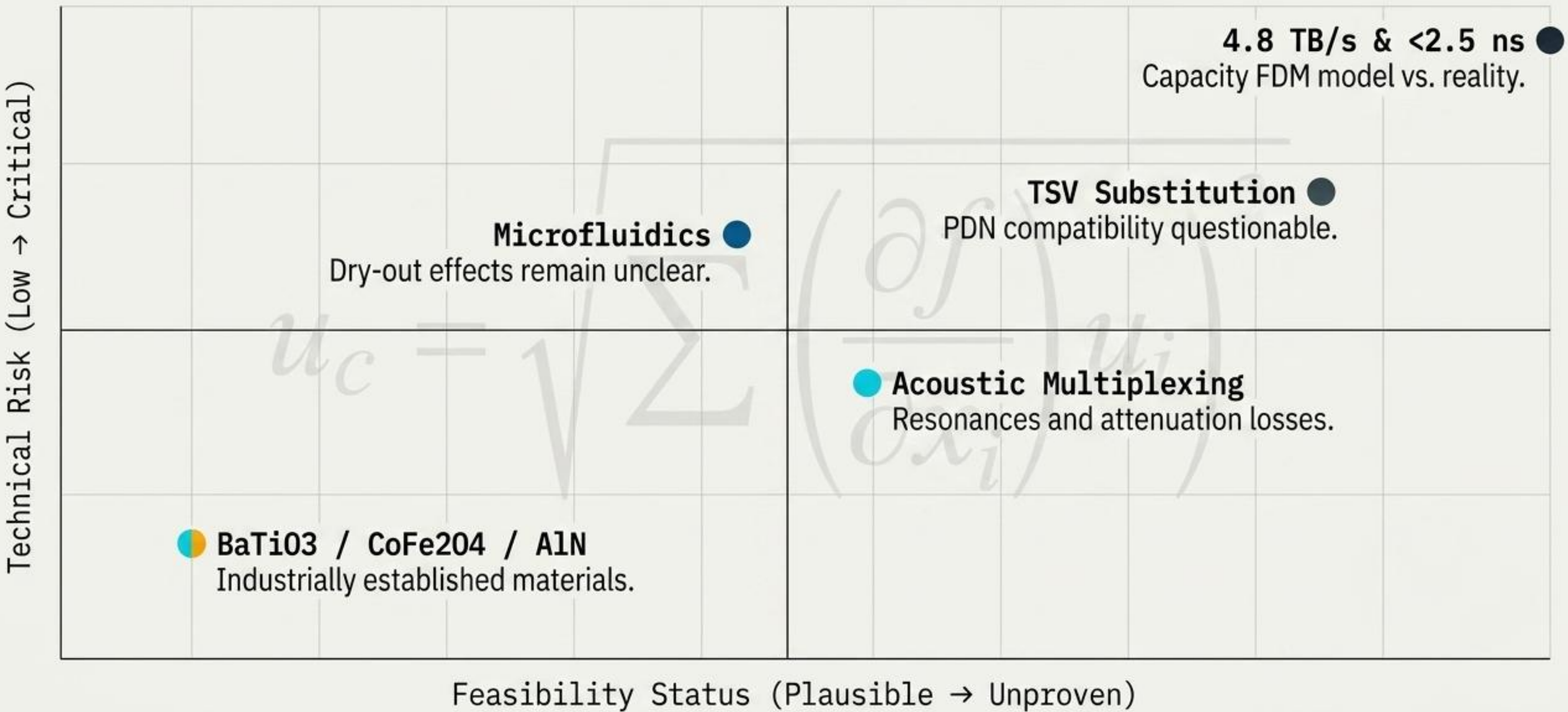
Murphy Model: $Y = \left(\frac{1 - e^{-DA}}{DA} \right)^2$

Disclaimer: Theoretical yield models assume successful substitution of metallic TSVs.



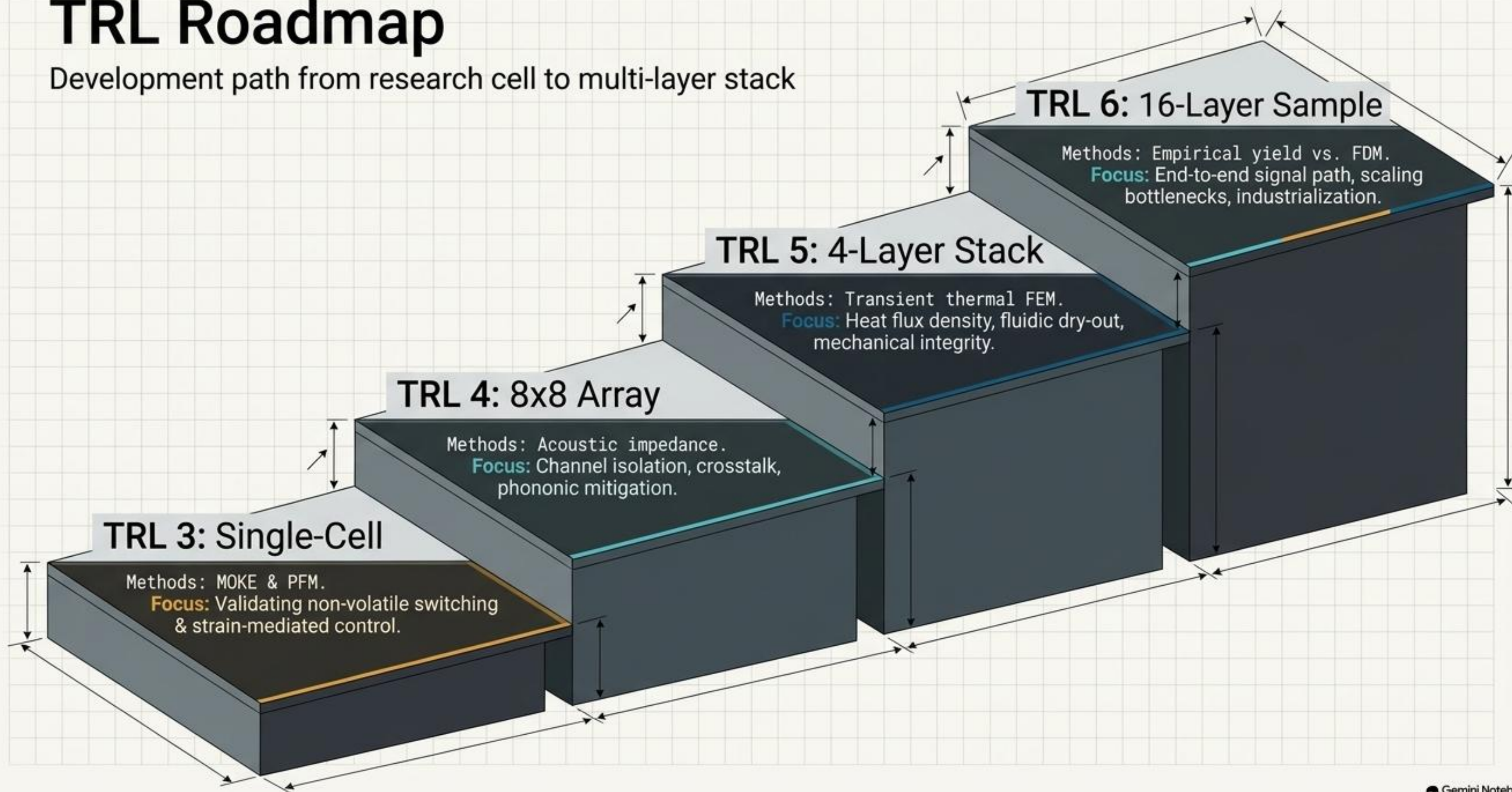
Risk & Uncertainty Analysis

Scientific mapping of architectural limitations.



TRL Roadmap

Development path from research cell to multi-layer stack



PCME-RAM-OSL (Open-Source License)

- **Permitted Usage:** Free to use, modify, test, benchmark, and integrate into commercial or private hardware.
- **Linux Compatibility:** Designed to integrate with Linux ecosystems, containerized environments, and OSD principles.
- **Modeling Framework:** Open access to OOA-SIM-V1 (Python feasibility framework).

Attribution Requirement:

Based on PCME-RAM, originally created by Emanuel Schaaf.
Contact: serviceblend@gmail.com

Building sovereign, open, and transparent AI hardware infrastructure.